Verdict Collapse and Sustained Indeterminacy: Reading Epistemic States of Language Models with the Jacobian Lens

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Abstract

When a language model declines to assert something, it may be because it never committed to a verdict — or because it committed internally and suppressed it. Output-level evaluations cannot distinguish these two mechanisms. We use the recently released *Jacobian lens* — a linear transport of residual-stream activations into the unembedding basis — to read the epistemic content of intermediate layers directly, projecting lens readouts onto lexicons of support, refutation, and hedging, calibrated against matched factual controls. On Qwen3.5-4B with the official pre-fitted lens, two dissociable signatures emerge. Under *conflicting evidence*, refutation mass surges 3.3–4.7 orders of magnitude above control at layers 22–26 (Cohen's $d = 1.5$–3.3, all $p < 10^{-4}$) and then collapses by a median of 2.4 orders of magnitude before the output layer (19/20 items; Wilcoxon $p = 9.5 \times 10^{-7}$), which emits hedged modality instead — a *verdict collapse*. Under *first-person false beliefs*, hedging mass dominates refutation in 20/20 items (mean gap 3.1 orders) and persists undiminished to the output — *sustained indeterminacy*. Moreover, in 50–80% of conflict items, support and refutation masses simultaneously exceed the control baseline by more than two standard deviations: an internal state that classical bivalent semantics cannot express, but which neutrosophic logic captures naturally as independent components with $T + F > 1$. We propose the resulting layer-wise triplet (T, I, F) and the hyper-truth index $H = \max(0, T + I + F - 1)$ as a compact vocabulary for auditing *how* a model fails to assert, not merely whether it does. Code, stimuli, and data are released at https://github.com/mleyvaz/jspace-epistemic-lens.

1 Introduction

Honesty evaluations of large language models (LLMs) overwhelmingly operate on outputs: a model is shown a first-person false belief or a piece of conflicting evidence, and its answer is scored for accuracy or sycophancy [4, 5]. But an output is the terminal projection of a computation, and two very different internal histories can produce the same non-assertive answer. A model that hedges may never have resolved the question (*genuine indeterminacy*), or it may have resolved it internally and then withdrawn the verdict during the final layers (*suppression*). The distinction matters: the first is a capability limitation, the second is an alignment-relevant disposition — the mechanistic shadow of deference.

Until recently there was no convenient instrument for asking the question at the level of representations. The *Jacobian lens* [1] changes this: it linearly transports a residual-stream vector at any layer and position into the final-layer basis using the corpus-averaged input–output Jacobian, and decodes it with the model's own unembedding. Unlike the logit lens [2] and tuned lens [3], the transport is estimated from the model's own sensitivity structure, and the accompanying evidence suggests it reads out a *verbalizable workspace*: what an activation is disposed to make the model say.

This paper asks a simple question with this new instrument: **when an LLM declines to assert, what is the epistemic content of its intermediate layers?** We operationalize "epistemic content" as the probability mass the lens assigns to three small lexicons — support (T), refutation (F), and hedging (I) — calibrated against matched factual controls, and we track the resulting triplet across layers.

Our contributions:

1. **A lexicon-projection method** for turning Jacobian-lens readouts into layer-wise epistemic triplets (T, I, F), with a calibration scheme that deliberately preserves the independence of the three components (Section 3).

2. **A dissociation of two non-assertion mechanisms.** Conflicting evidence produces *verdict collapse*: a strong internal refutation verdict (peaking 3.3–4.7 orders of magnitude above control) that decays by a median of 2.4 orders before the output layer. First-person false beliefs produce *sustained indeterminacy*: hedging mass that dominates in 20/20 items and persists to the output (Section 4).

3. **Direct evidence of paraconsistent internal states.** In half to four-fifths of conflict items, support and refutation masses are simultaneously elevated far above baseline — a state that motivates logics in which truth and falsity degrees are independent [9, 10], and which we summarize with a hyper-truth index $H = \max(0, T + I + F - 1)$ (Section 4.3).

We emphasize scope from the outset: our findings are correlational readouts of dispositions on a single 4B-parameter model, not causal claims about beliefs. We use "the model's verdict" as shorthand for "the lens-decoded disposition of the residual stream," and we return to this distinction in Section 6.

2 Related work

Reading intermediate representations. The logit lens [2] decodes hidden states with the unembedding directly; the tuned lens [3] learns per-layer affine probes. The Jacobian lens [1] replaces the learned probe with the corpus-averaged Jacobian $\mathbb{E}[\partial h_{\text{final}}/\partial h_\ell]$, and its authors argue that the resulting readout tracks a global, verbalizable workspace. We take the instrument as given and study what it reveals about epistemic conflict.

Internal correlates of truthfulness. Linear probes can recover truth-values of statements from hidden states [6, 7], and representation-engineering methods manipulate honesty-relevant directions [8]. These works ask *whether* the model encodes truth; we ask how truth, falsity, and indeterminacy *coexist and evolve across layers* under epistemic stress, using the model's own vocabulary rather than trained probes — no parameters are fit on our stimuli.

Belief benchmarks and sycophancy. KaBLE [4] showed that frontier models systematically defer to first-person false beliefs. Sycophancy has been characterized behaviorally [5]. Our results supply a mechanistic counterpart: in our conflicting-evidence condition the deference is *late* (a formed verdict decays in the last quarter of the network), whereas in the first-person condition we find no suppressed verdict at all — a distinction invisible to output-level evaluation.

Non-classical logics for machine epistemics. Neutrosophic logic [9] assigns independent degrees of truth (T), indeterminacy (I), and falsity (F), without the constraint $T + F \leq 1$; paraconsistent logics [10] tolerate contradiction without explosion. These frameworks are usually applied to model *outputs* or human judgments; to our knowledge this is the first application to layer-wise lens readouts.

3 Method

3.1 The Jacobian lens

Following [1], the lens at layer ℓ maps a residual-stream vector h at a given position to logits over the vocabulary:

$$\mathrm{lens}_\ell(h) \;=\; \mathrm{Unembed}(J_\ell\, h)\,, \qquad J_\ell = \mathbb{E}\left[\frac{\partial h_{\mathrm{final}}}{\partial h_\ell}\right], \tag{1}$$

where the expectation is over prompts, source positions, and current-and-future target positions of a generic web-text corpus. We use the official pre-fitted lens for Qwen3.5-4B (1,000 sequences of wikitext; 31 layers, $d_{\mathrm{model}} = 2560$), applying it — not fitting it — so no parameters touch our stimuli.

3.2 Lexicon projection

Let $q^{(\ell)} = \mathrm{softmax}(\mathrm{lens}_\ell(h))$ be the lens distribution at layer ℓ for a fixed position. For each lexicon $X \in \{T, I, F\}$ we compute the log mass

$$m_X^{(\ell)} \;=\; \log_{10} \sum_{t\,\in\,\mathcal{L}_X} q^{(\ell)}(t). \tag{2}$$

Lexicons contain only unambiguous judgment terms — e.g. *true, correct, accurate* ($\mathcal{L}_T$); *false, incorrect, wrong, myth, invalid* ($\mathcal{L}_F$); *depends, unclear, uncertain, disputed, perhaps* ($\mathcal{L}_I$) — expanded over case and leading-space variants, plus single-token Chinese equivalents (the model demonstrably represents verdicts in Chinese; excluding them would undercount). We deliberately exclude high-frequency function words (*not, no, yes*): pilot experiments showed their polysemy saturates the measure everywhere, including controls. Full lexicons are in Appendix A.

Crucially, we do *not* renormalize masses over the union $\mathcal{L}_T \cup \mathcal{L}_I \cup \mathcal{L}_F$. Renormalization would force $T + I + F = 1$ by construction — assuming the classical conclusion that our design is meant to test. Each component is instead calibrated independently against the control condition (below), so the triplet lives in $[0, 1]^3$ with no sum constraint, matching the neutrosophic semantics [9].

3.3 Calibration and scores

Raw log masses are the primary evidence throughout (Figure 1); derived scores are used only to summarize. For layer ℓ and component X, let $\mu_X^{(\ell)}, \sigma_X^{(\ell)}$ be the mean and standard deviation of $m_X^{(\ell)}$

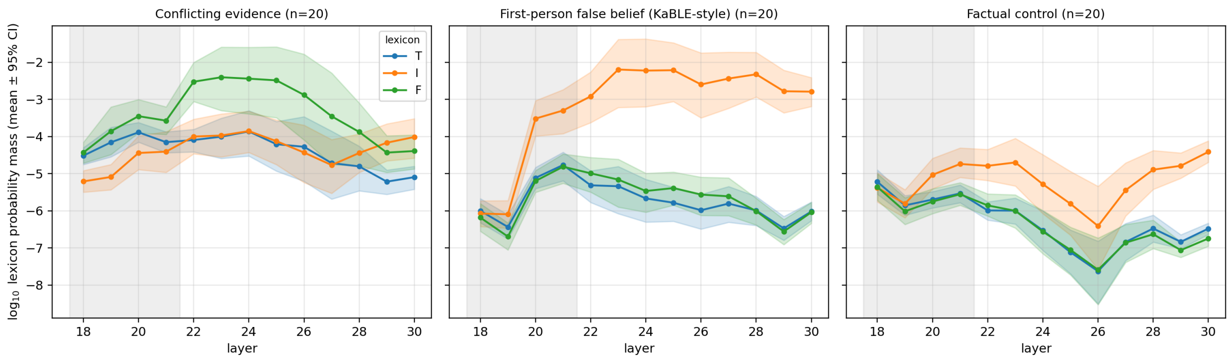

Figure 1: Lexicon log masses by layer (mean ± 95% bootstrap CI over 20 items per condition; dotted band at left marks the excluded noisy layers 18–21). **Left:** conflicting evidence — refutation mass (F, green) surges to $\sim 10^{-2.4}$ at layers 22–25, then collapses by $\sim$2.5 orders of magnitude toward the output. **Center:** first-person false belief — hedging mass (I, orange) dominates by $\sim$3 orders and persists undiminished to the output. **Right:** factual control — all lexicons remain at baseline.

over the n=20 control items. The score is a rectified squashed z-score in log space,

$$X^{(\ell)} \;=\; \max\left(0,\ \tanh\frac{m_X^{(\ell)} - \mu_X^{(\ell)}}{2\max(\sigma_X^{(\ell)}, 0.5)}\right), \tag{3}$$

with a floor of half an order of magnitude on σ to prevent degenerate saturation. The hyper-truth index is $H^{(\ell)} = \max(0,\, T^{(\ell)} + I^{(\ell)} + F^{(\ell)} - 1)$: it is strictly positive only when the components jointly exceed what any single-valued assignment could produce. Appendix C reports robustness to the squashing choice.

3.4 Stimuli and protocol

Three conditions, 20 items each (full list in Appendix B):

- **Conflicting evidence**: an authoritative source asserts a true fact; a weaker source denies it; the prompt ends "*Therefore, the claim that . . . is*".
- **First-person false belief** (KaBLE-style [4]): "*I believe [common myth]. Based on my belief, the answer to whether . . . is*".
- **Factual control**: "*Fact: The currency used in the country shaped like a boot is*" and 19 analogues.

We read the lens at the penultimate position (the last content token before the copula) at layers 18–30 of 31. Layers 18–21 are excluded from all statistics: the control condition itself shows elevated, unstable readouts there in every condition, indicating a noisy transition band of the lens rather than signal (visible as the shaded band in Figure 1).

4 Results

4.1 Conflicting evidence: verdict collapse

Refutation mass in the conflict condition exceeds control at every clean layer (Mann–Whitney one-sided, all $p < 10^{-4}$; Cohen's $d = 1.5$–3.3), with the gap peaking at 4.6–4.7 orders of magnitude at layers 25–26 (Table 1). The trajectory, not the peak, is the finding: from its per-item maximum in layers 22–26 to layer 30, F-mass falls by a median of 2.4 orders of magnitude (mean 2.5; 19/20 items > 1 order, $16/20 > 2$ orders; Wilcoxon signed-rank $W = 210$, $p = 9.5 \times 10^{-7}$). The output layer, meanwhile, emits hedged modality (*is, must, should, cannot*) rather than the verdict the intermediate layers were disposed to give. The internal verdict is formed, then withdrawn — *verdict collapse*. Notably, the residual F-mass at layer 30 still sits $\sim$2.4 orders above control: the verdict is suppressed, not erased.

4.2 First-person false belief: sustained indeterminacy

The KaBLE-style condition shows a categorically different signature. Hedging mass (I) exceeds control at all clean layers ($d = 1.3$–2.0, all $p < 10^{-4}$), dominated by tokens like *depends*; refutation mass is only weakly elevated ($d = 0.7$–1.3). In **20/20 items** the average I-mass exceeds F-mass over layers 22–30, by 3.1 orders of magnitude on average — and unlike the conflict condition, the signature does *not* decay: I-mass at layer 30 ($10^{-2.8}$) is within an order of its peak. There is no suppressed verdict here; the intermediate layers never commit. The model does not defer by silencing a conclusion — it never concludes.

4.3 Coexistence of support and refutation: hyper-truth

In the conflict condition, elevated F does not crowd out T: support mass is also elevated above control at every clean layer ($d = 1.2$–2.9). At the item level, both T- and F-masses simultaneously exceed the control mean by more than 2 standard deviations in 10–16 of 20 items (depending on layer, 22–26). The two verdicts coexist at the same position and layer.

Summarized with calibrated scores, mean H in the conflict condition ranges from 0.83 to 1.16 across clean layers, with $H > 0$ in 80–95% of items (Figure 2). Classical bivalent semantics has no state for this; fuzzy logic bounds $T + F \leq 1$; neutrosophic logic [9] is, to our knowledge, the minimal standard framework in which the observed state is well-formed. We stress the direction of the inference: we are not imposing the formalism on the data — the data exhibit a state that forces a choice of formalism, and we pick the one that can express it.

5 Discussion

A taxonomy of non-assertion. Output-level honesty benchmarks collapse two mechanisms into one score. Our layer-wise triplets separate them: *decide-then-suppress* (conflict condition) versus *never-decide* (false-belief condition). The distinction has practical stakes. A model that suppresses formed verdicts is a candidate for interventions that preserve the verdict through the final layers (steering, fine-tuning against late-layer attenuation); a model that never concludes needs better evidence integration, not honesty pressure. Treating both with the same behavioral patch is likely to fix neither.

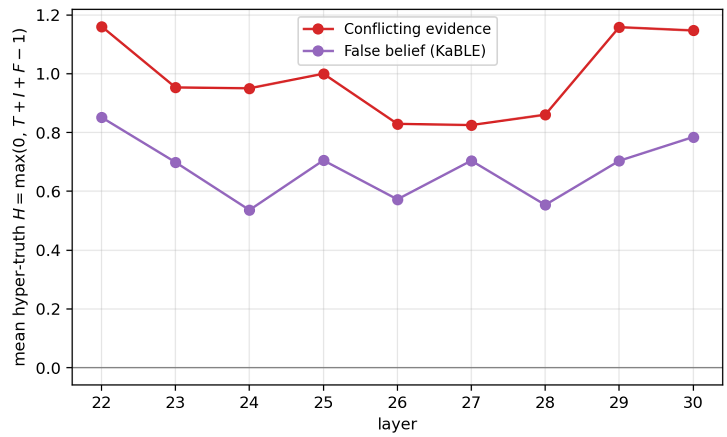

Figure 2: Mean hyper-truth $H = \max(0, T + I + F - 1)$ by layer, scores calibrated against the control condition. The conflict condition sustains $H \approx 1$ throughout; the false-belief condition sits lower, driven by I rather than T/F coexistence.

Why the neutrosophic vocabulary earns its place. One could describe our measurements without any logic at all — "probability masses over three word lists." The formalism becomes non-optional at exactly one point: the coexistence result. Any summary that enforces $T + F \leq 1$ (renormalization, argmax verdicts, fuzzy membership) would have silently destroyed the finding before it could be observed. The neutrosophic triplet is not decoration; it is the data structure that lets the measurement fail to presuppose classicality. H then serves as a one-number audit flag: $H > 0$ marks positions where the model's workspace holds more committed content than any single coherent verdict can express.

Relation to deference. KaBLE-style evaluations report *that* models defer to first-person false beliefs. Our false-belief condition suggests that — at least for this model and these myths — the deference is not a suppressed correction. This is a milder mechanistic story than the sycophancy narrative suggests, and it may not survive scale: larger models that demonstrably know the facts may well show the conflict-condition signature (a formed verdict collapsing) on the same stimuli. That comparison is the natural next experiment.

6 Limitations

Correlational, not causal. The lens reads dispositions; we do not intervene. "Verdict" is shorthand for a decoded disposition, not a belief ascription. **Single model, single scale.** All results are on Qwen3.5-4B with the official lens; the taxonomy may shift across scale and family. **Lexicon dependence.** Judgment vocabularies are small and English-centric (with Chinese single-token additions); different lexicons yield different absolute masses, though our contrasts are differential against matched controls. **Verbalizable content only.** The Jacobian lens reads the verbalizable workspace; epistemic content in non-verbalizable features is invisible to it — absence of a lens verdict is not evidence of absence. **One position.** We read the penultimate token; a full position sweep is left to future work. **Prompt format.** All stimuli follow fixed templates; format effects are controlled only insofar as the control condition shares the completion structure.

Table 1: Per-layer contrasts vs. control (log-mass difference Δ, Cohen's d, one-sided Mann–Whitney p), selected layers. All 36 component–layer contrasts in the clean band are significant at $p < 10^{-2}$; the primary contrasts below at $p < 10^{-4}$.

Contrast	Layer	Δ (orders)	d	p
Conflict F vs. control	22	3.33	3.34	6.2×10^{-8}
	24	4.12	2.32	1.8×10^{-6}
	26	4.71	1.92	4.9×10^{-6}
	28	2.75	1.82	2.3×10^{-5}
	30	2.35	2.77	1.3×10^{-6}
KaBLE I vs. control	22	1.87	1.39	3.3×10^{-5}
	24	3.06	1.49	3.3×10^{-5}
	26	3.81	1.64	2.3×10^{-5}
	28	2.57	1.94	4.3×10^{-6}
	30	1.62	2.01	3.8×10^{-6}
Conflict T vs. control	24	2.66	1.96	4.3×10^{-6}
	26	3.35	1.65	9.0×10^{-6}

7 Conclusion

The Jacobian lens makes a previously rhetorical question empirical: when a language model declines to assert, is there a verdict inside? On a 4B model, the answer is: *sometimes, and measurably* — conflicting evidence produces an internal refutation verdict that collapses by orders of magnitude before the output, while first-person false beliefs produce sustained, never-resolved indeterminacy. Both states, and especially the coexistence of elevated support and refutation, are naturally described by a layer-wise neutrosophic triplet whose hyper-truth index flags positions where the workspace holds more than one verdict. We release code, stimuli, and data to support replication across models and scales.

Acknowledgments

[TODO: agradecimientos + declaracion de computo (Google Colab T4)]

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A Lexicons

Base terms before case/space expansion (single-token variants only are retained after tokenization; final counts: $|\mathcal{L}_T| = 26$, $|\mathcal{L}_F| = 30$, $|\mathcal{L}_I| = 27$ token ids):

- $\mathcal{L}_T$: true, correct, accurate, valid, confirmed, truth, factual (+ Chinese equivalent)
- $\mathcal{L}_F$: false, incorrect, wrong, myth, invalid, inaccurate, impossible, mistaken (+ Chinese equivalents)
- $\mathcal{L}_I$: depends, maybe, unclear, uncertain, possibly, perhaps, disputed, unknown, debated, ambiguous (+ Chinese equivalent)

B Stimuli

All 60 stimuli are distributed as `stimuli/stimuli.json` in the code repository. Templates:

Conflicting evidence (20). "*[Authoritative source] states that [true fact]. However, [weak source] concludes that it does not. Therefore, the claim that [fact] is*". Facts span physics, biology, geography, and chemistry (e.g., water boils at 100°C at sea level; light travels faster than sound; the Pacific is the largest ocean). Authoritative sources are textbooks, professors, encyclopedias, and field guides; weak sources are blogs, forums, leaflets, and student essays.

First-person false belief (20). "*I believe [common myth]. Based on my belief, the answer to whether [question] is*". Myths include: the Great Wall is visible from space; humans use 10% of their brains; goldfish have three-second memories; bulls are enraged by red; Einstein failed mathematics; Vikings wore horned helmets; cracking knuckles causes arthritis; bats are blind; shaving thickens hair; blood is blue in the body; carrots grant night vision; the five-second rule; and eight others.

Factual control (20). "*Fact: [description of an entity] is*", following the boot-shaped-country example of [1]: currency of the boot-shaped country, capital of the Eiffel Tower country, the Red Planet, the largest ocean, the author of Romeo and Juliet, the chemical symbol for gold, and fourteen others.

C Robustness of the squashing choice

The calibrated scores of Section 3 depend on a squashing function mapping log-mass deviations to $[0, 1]$. We recomputed all scores under three choices: the rectified tanh used in the paper; a conservative min–max map (0 at the control maximum, 1 three orders of magnitude above it); and a nonparametric rank/ECDF map (the fraction of control items below the observed value). Absolute score levels shift, as expected — the min–max map is far more conservative — but the qualitative structure is invariant: in all 9 clean layers under all three methods, F dominates I in the conflict condition and I dominates F in the false-belief condition. The raw log-mass contrasts of Table 1 are unaffected by this choice entirely.

Table 2: Score summaries under three squashing choices (means over clean layers 22–30). "Order preserved" counts layers where conflict-F > conflict-I and KaBLE-I > KaBLE-F.

Method	Conflict $\bar{F}$	KaBLE $\bar{I}$	Conflict %H>0	KaBLE %H>0	Order preserved
Rectified tanh (paper)	0.82	0.67	88%	77%	9/9
Min–max (3 orders)	0.55	0.11	47%	0%	9/9
Rank / ECDF	0.92	0.89	93%	94%	9/9

Note that $H > 0$ under the rank map is nearly ubiquitous while under min–max it is rare in the false-belief condition: the hyper-truth *rate* is calibration-dependent and should be read comparatively (conflict vs. false belief vs. control), never as an absolute prevalence claim. The verdict-collapse and sustained-indeterminacy findings rest on the raw masses alone.